

RK>

Applied ML & Document Intelligence

RFID Localization • Semantic Search • Extractive QA • OCR • Data Pipelines

Applied ML • NLP • RFID

400+

tagged products

RFID localization

8

office zones

classification targets

2K+

labeled samples

simulation
generated samples

20%

lower ingestion
latency

4-stream + schema
redesign

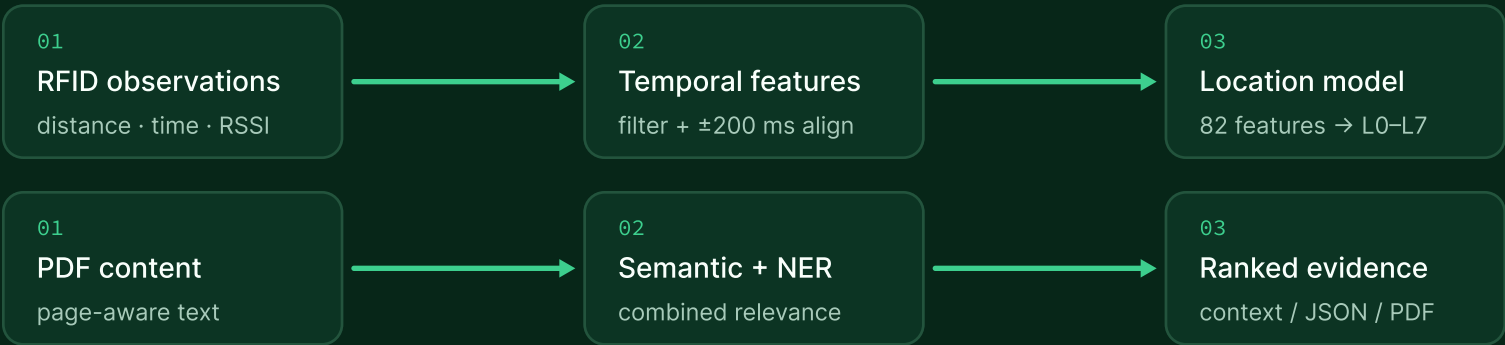
ENGINEERING NARRATIVE

Built two applied Python/ML systems: RFID localization and semantic document intelligence.

RFID work modeled signal behavior, temporal feature engineering, and eight-class localization; document work combined semantic similarity, custom NER, extractive QA, OCR, highlighting, and structured outputs.

TWO PRIMARY SYSTEMS

Independent applied-ML paths for physical localization and document understanding.



ENGINEERING SCOPE

- RFID Localization
- Signal Modeling
- Semantic Search
- Extractive QA
- OCR / Parsing
- Data Pipelines

CORE TECHNOLOGIES

Python • pandas • TensorFlow / Keras • NLTK • spaCy • Haystack • FAISS • PostgreSQL
PyPDF2 • pdfplumber • PyMuPDF • pdf2image • pytesseract • pygame

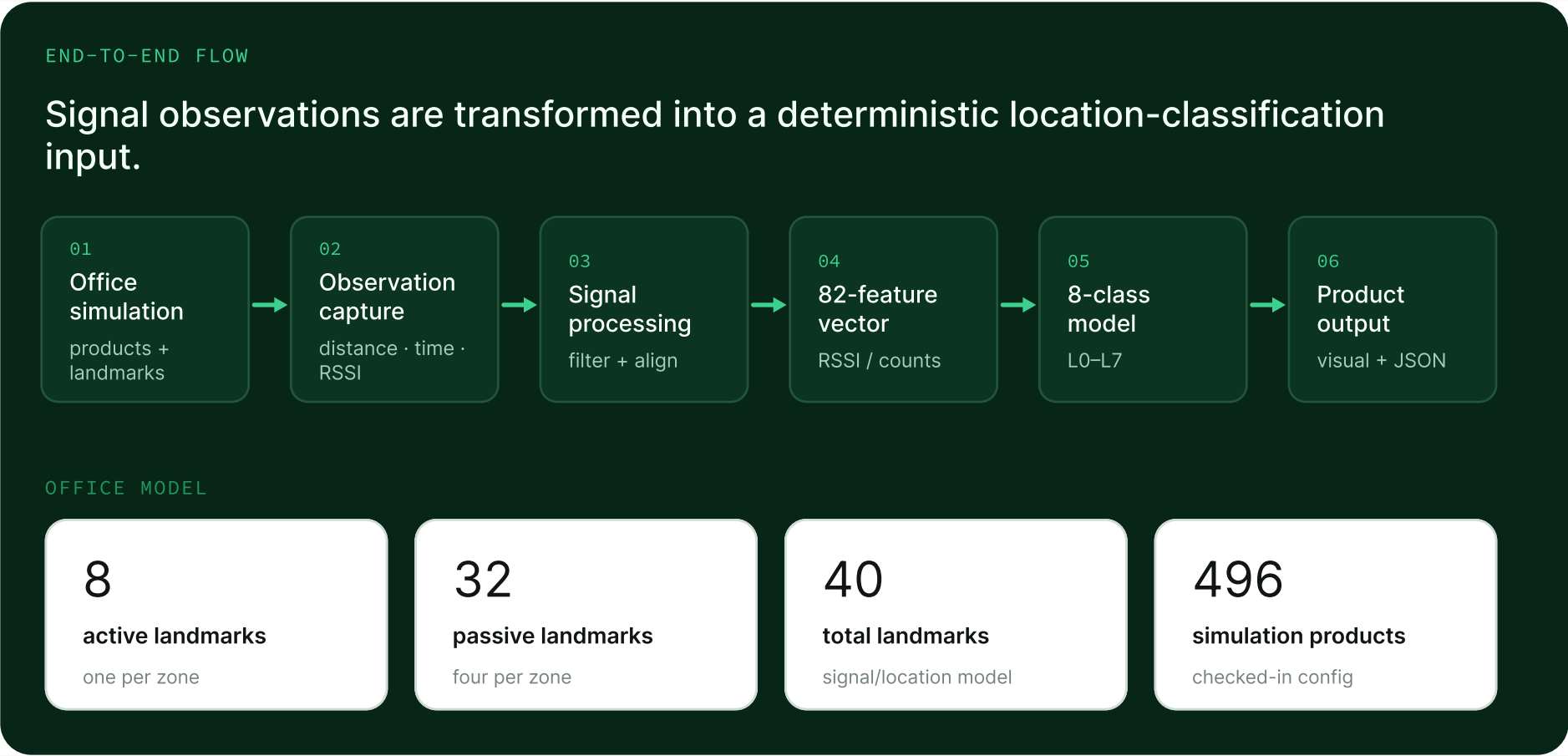
RFID Localization Architecture

From simulated signal observations to product-level location classes.

SYSTEM

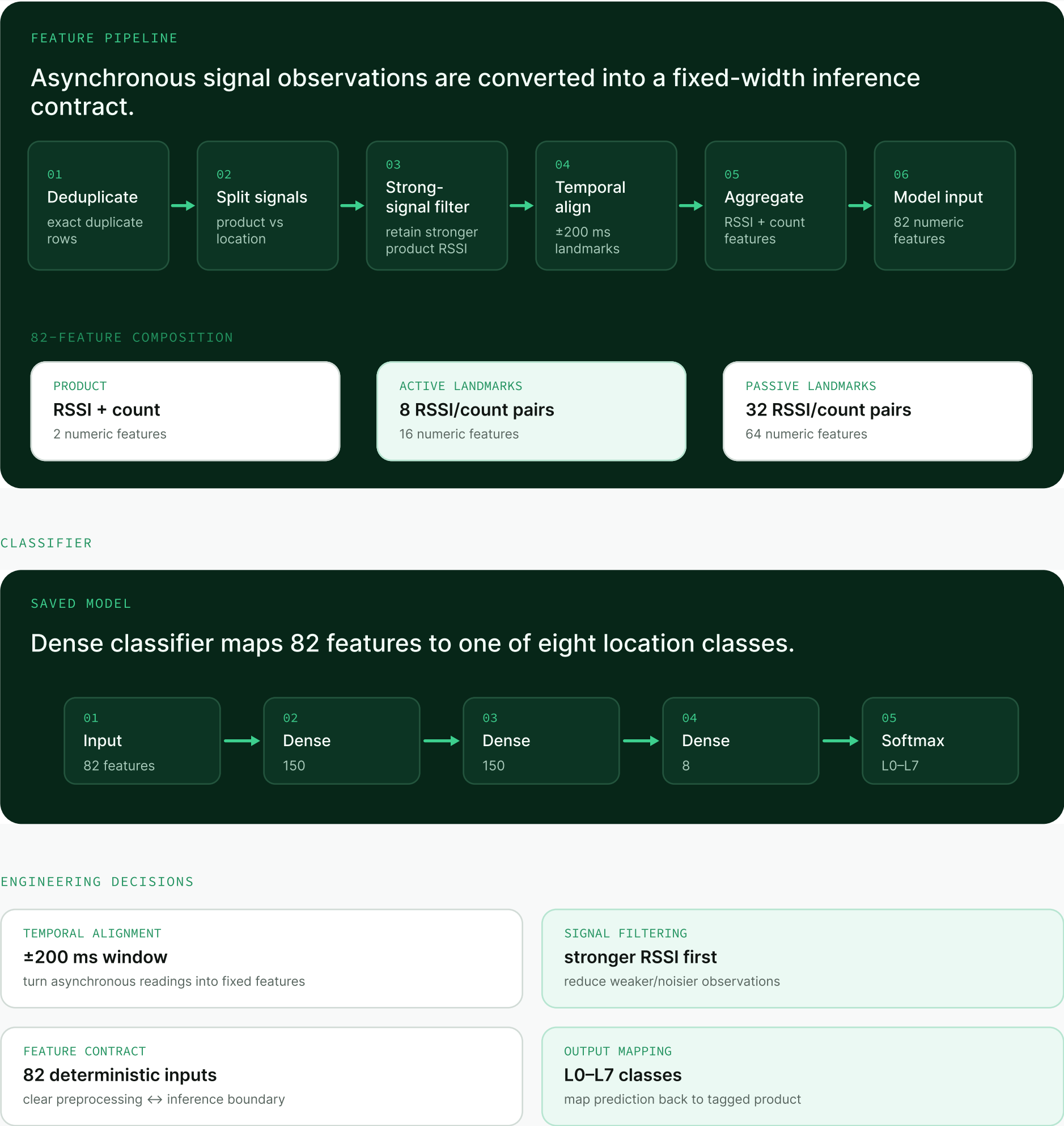
RFID indoor localization

Simulated office observations → RSSI modeling → temporal alignment → fixed-width features → eight-class location prediction.



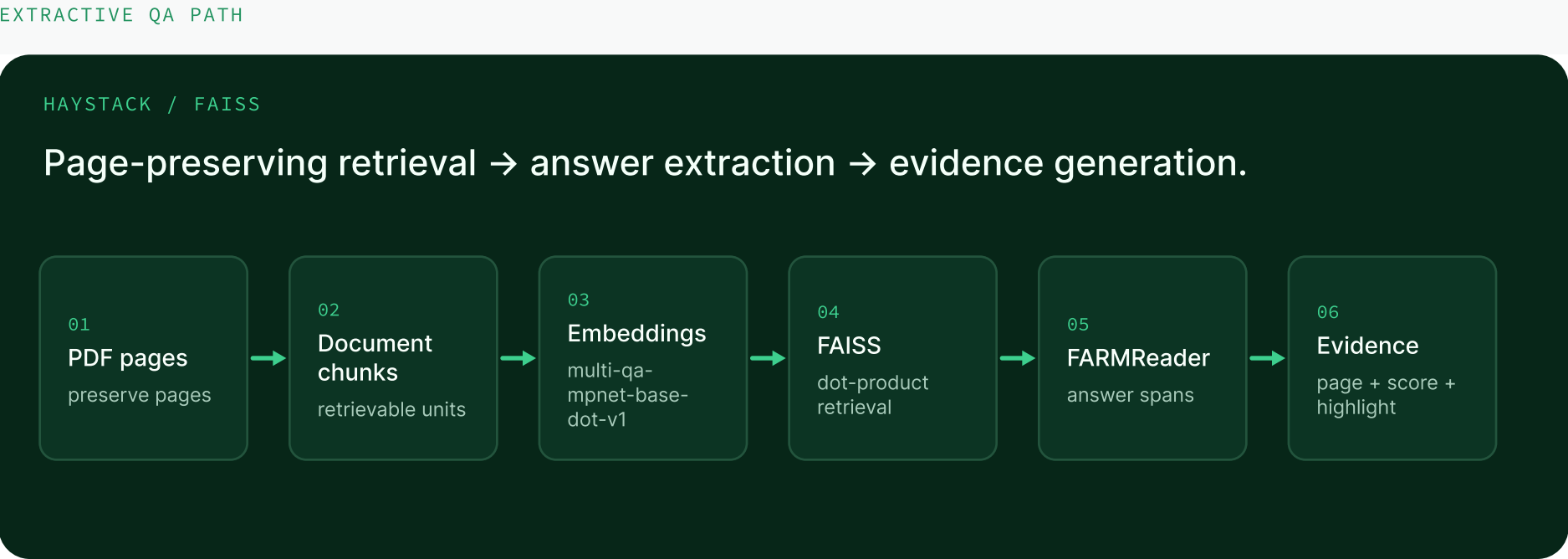
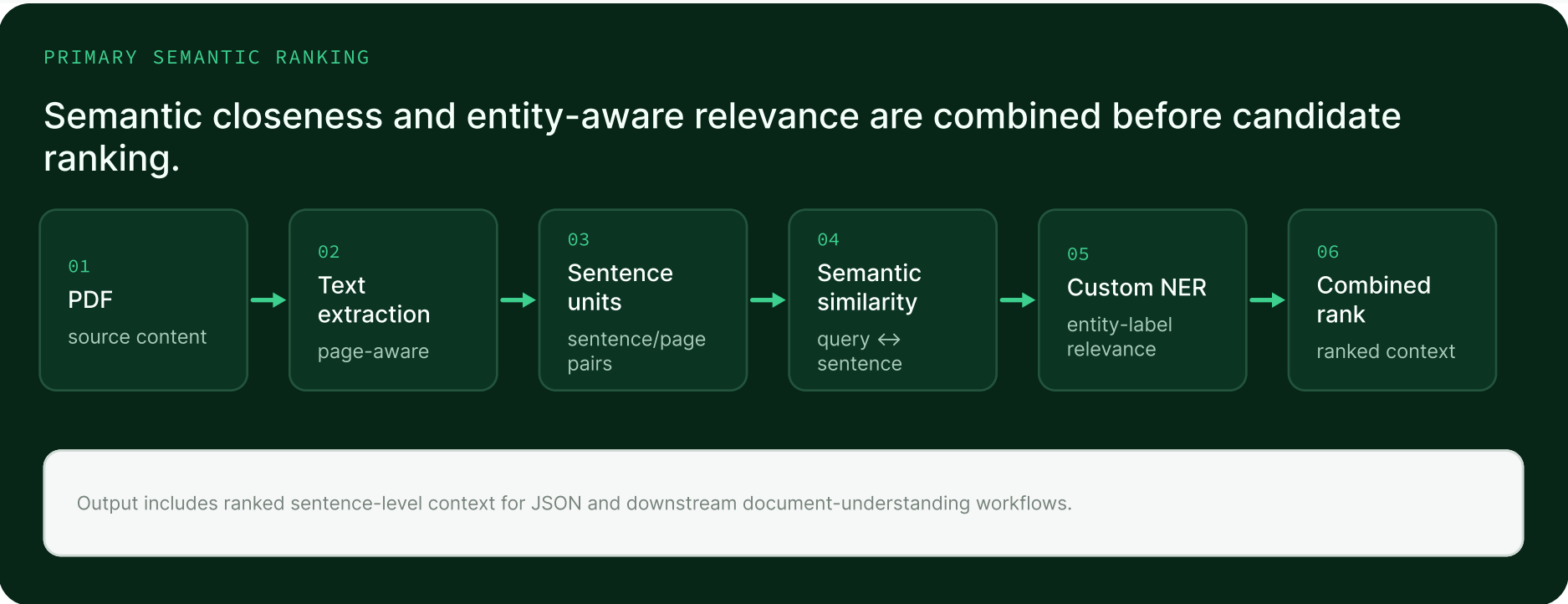
Feature Engineering & Classifier

A deterministic RSSI/count contract feeding an eight-class neural classifier.

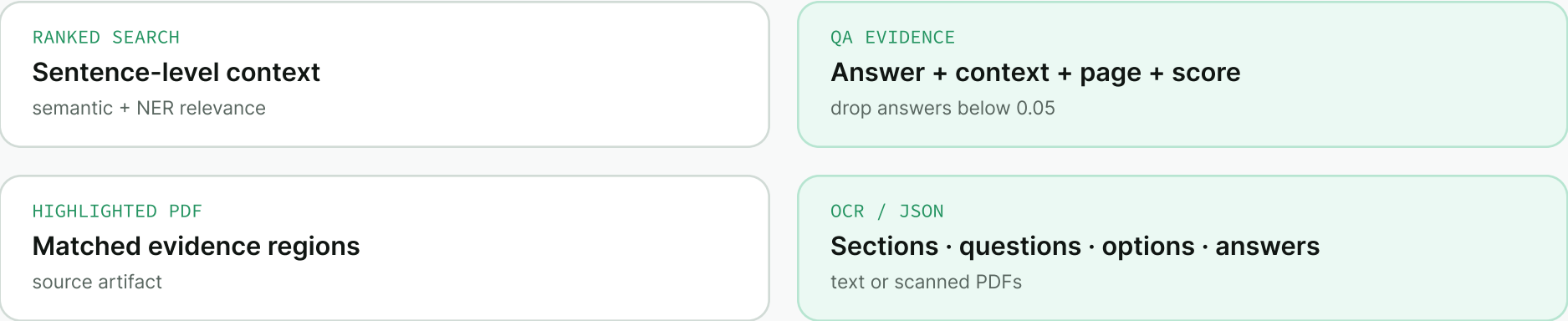


Document Intelligence Systems

Semantic ranking, extractive QA, evidence highlighting, OCR, and structured output.



DOCUMENT OUTPUTS



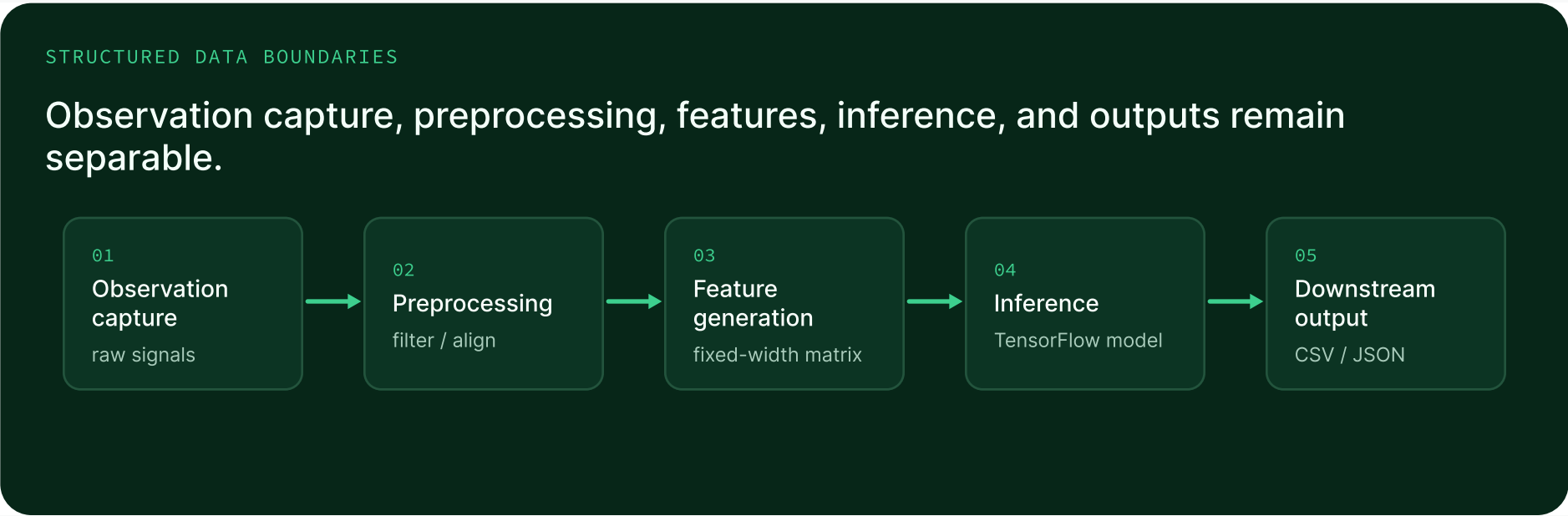
Ingestion & Data Engineering

Parallelized processing, normalized schemas, and inspectable ML data boundaries.

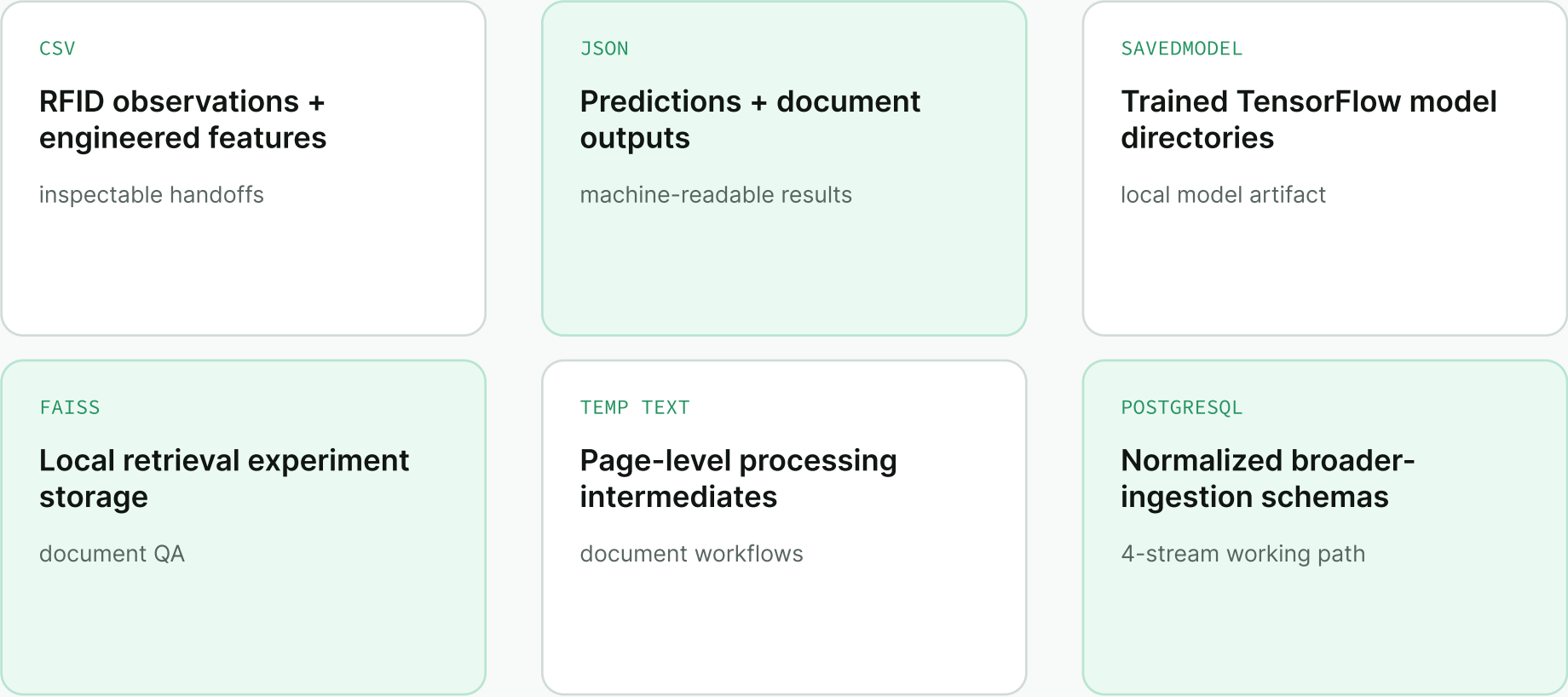
INGESTION PATH

4 parallel streams + normalized PostgreSQL

The broader implementation parallelized four ingestion streams and redesigned normalized PostgreSQL schemas, reducing ingestion latency by 20%.



DATA & PERSISTENCE



Repository snapshot uses local CSV/JSON handoffs; PostgreSQL-backed ingestion belongs to the broader confirmed implementation and is not represented by the local-file-only snapshot.

Results, Decisions & Scope

Confirmed scale, engineering trade-offs, and publication-safe boundaries.

CONFIRMED RESULTS

SYSTEM	RESULT	FRAMING
RFID localization	400+ tagged products across 8 zones	confirmed project scale
Signal model	40 landmark tags · 2K+ samples	simulation-generated samples
Feature contract	82 numeric RSSI/count inputs	saved classifier
Ingestion	20% lower latency	4 streams + normalized PostgreSQL
Simulation snapshot	496 products	checked-in configuration

ENGINEERING DECISIONS

SIMULATION repeatable signal generation avoid hardware dependency every iteration	TEMPORAL ALIGNMENT ±200 ms association window make asynchronous signals explicit
STRONG-SIGNAL FILTER prioritize stronger RSSI reduce weaker/noisier observations	ENTITY-AWARE RANKING semantic + NER relevance combine meaning with entity information

SCOPE BOUNDARY

Public representation: applied ML, NLP, RFID localization, document processing, and data-pipeline engineering. Do not add web SaaS, frontend, auth, multi-tenancy, cloud deployment, Kubernetes, CI/CD, uptime, traffic, API-throughput, revenue, or adoption claims without a separate confirmed source.
TECHNICAL STACK Python · pandas · TensorFlow / Keras · NLTK · spaCy · Haystack · FAISS · PostgreSQL PyPDF2 · pdfplumber · PyMuPDF · pdf2image · pytesseract · pygame